

NETWORKED CONTROL

Prof. Marcello Farina

NAME/FAMILY NAME..... SOLUTIONS.....
PERSONAL CODE.....
SIGNATURE.....
DATE..... 26/06/23.....

Notes:

- The solutions must be written in the pages of this leaflet.
- At the end of the exam only this file is accepted, **without any additional page**.
- No notes or books are accepted, neither in paper nor electronic form.
- Clarity and precision will be matter of evaluation.

$$1) \quad \dot{x} = -x + 2u$$

$$y = 2x$$

$$A = -1, B = 2$$

$$C = 2, D = 0$$

A. a. . EIGENVALUES: $\lambda = -1$

. MODE: e^{-t}

. THE SYSTEM IS ASYMPTOTICALLY STABLE (IN OPEN LOOP)

$$b. u = Ky = 2Kx \rightarrow \dot{x} = (-1 + 4K)x$$

THE CLOSED-LOOP SYSTEM IS ASYMPTOTICALLY STABLE IFF

$$-1 + 4K < 0 \Rightarrow K < \frac{1}{4}$$

$$B. h = \log_e 2 \rightarrow$$

$$F = e^{-\log_e 2} = \frac{1}{2}$$

$$G = \frac{b}{a} (e^{ah} - 1) = -2 \left(\frac{1}{2} - 1 \right) = 1$$

$$H = C = 2$$

$$\Rightarrow \begin{cases} x_{k+1} = \frac{1}{2} x_k + u_k \\ y_k = 2x_k \end{cases}$$

$$\text{IN CLOSED LOOP } u_k = Ky_k = 2Kx_k$$

$$\rightarrow x_{k+1} = \left(\frac{1}{2} + 2K \right) x_k$$

$$\left| \frac{1}{2} + 2K \right| < 1 \quad \begin{cases} \frac{1}{2} + 2K < 1 \\ \frac{1}{2} + 2K > -1 \end{cases} \Leftrightarrow \boxed{-\frac{3}{4} < K < \frac{1}{4}}$$

$$C. a. \underline{F} = \begin{bmatrix} e^{A(h-z)} + \Gamma(h-z)BK C & e^{A(h-z)} \Gamma(h-z)BK \\ C & 0 \end{bmatrix}$$

$$h = \log_e 2 \rightarrow$$

$$K = -\frac{1}{2}$$

$$e^{Ah} = \frac{1}{2} \quad \Gamma(h-z) = \frac{1}{A} (e^{A(h-z)} - I) = - \left(e^{-\log_e 2 + z} - 1 \right)$$

$$= - \left(\frac{1}{2} e^z - 1 \right)$$

$$e^{A(h-z)} = \frac{1}{2} e^z, \quad \Gamma(z) = - (e^{-z} - 1)$$

$$\underline{F}(h, z) = \begin{bmatrix} \frac{1}{2} - \left(\frac{1}{2} e^z - 1 \right) \left(-\frac{1}{2} \right) \cdot 4 & -\frac{1}{2} e^z (e^{-z} - 1) \left(-\frac{1}{2} \right) 2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -\frac{3}{2} + e^z & \frac{1}{2} (1 - e^z) \\ 2 & 0 \end{bmatrix}$$

$$e^z = T \Rightarrow \underline{F} = \begin{bmatrix} -\frac{3}{2} + T & \frac{1}{2} (1 - T) \\ 2 & 0 \end{bmatrix}$$

$$p_2(\lambda) = \det \begin{bmatrix} \lambda - (\tau - \frac{3}{2}) & -\frac{1}{2}(1-\tau) \\ -2 & \lambda \end{bmatrix} = \lambda^2 - (\tau - \frac{3}{2})\lambda - (1-\tau)$$

THE NCS IS ASYMPTOTICALLY STABLE IFF

$$\begin{cases} |\tau - 1| < 1 \\ |\tau - \frac{3}{2}| < 1 + \tau - 1 \end{cases} \Rightarrow \begin{cases} -1 < \tau - 1 < 1 \\ -\tau < \tau - \frac{3}{2} < \tau \end{cases} \quad \begin{matrix} 0 < \tau < 2 \\ \Leftrightarrow \tau > \frac{3}{2} \end{matrix}$$

$$\Rightarrow \underbrace{\frac{3}{4} < e^{\tau} < 2}_{\forall \tau} \quad \boxed{\tau < \log_e 2}$$

$$b. \mathbb{I}_1 = \begin{bmatrix} -\frac{3}{2} + e^{\tau} & \frac{1}{2}(1 - e^{\tau}) \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -\frac{3}{2} + 1.5 & \frac{1}{2}(1 - 1.5) \\ 2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -\frac{1}{4} \\ 2 & 0 \end{bmatrix}$$

$$\mathbb{I}_0 = \begin{bmatrix} F & GK \\ 0 & I \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{bmatrix}$$

THE 3-STEPS TRANSITION MATRIX IS

$$\mathbb{I}_3 = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -\frac{1}{4} \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & -\frac{3}{4} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -\frac{1}{4} \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -\frac{3}{2} & -\frac{1}{16} \\ 2 & 0 \end{bmatrix}$$

$$p_{\mathbb{I}_3}(\lambda) = \det \begin{bmatrix} \lambda + \frac{3}{2} & \frac{1}{16} \\ -2 & \lambda \end{bmatrix} = \lambda^2 + \frac{3}{2}\lambda + \frac{1}{8}$$

$$\left. \begin{matrix} \frac{1}{8} < 1 \\ \frac{3}{2} < 1 + \frac{1}{8} \end{matrix} \right\} \text{Not as stable}$$

